

Machine learning is a branch of artificial intelligence that focuses on building systems that can learn from data. Instead of relying solely on explicit programming, machine learning models identify patterns in large datasets and use those patterns to make predictions or decisions.

Supervised learning, unsupervised learning, and reinforcement learning are three common types of machine learning techniques. In supervised learning, algorithms are trained using labeled data where the correct outputs are known. In unsupervised learning, the model analyzes unlabeled data to find hidden structures or patterns. Reinforcement learning involves training an agent through trial and error to maximize a reward signal.

Machine learning is widely used in applications such as recommendation systems, fraud detection, natural language processing, and image recognition. As the availability of data continues to grow, machine learning techniques are becoming increasingly important for solving complex problems in many industries.